

Gabriel Wan

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EDUCATION

Bachelor of Computing in Computer Science (Honours), GPA: 4.43/5.00

Aug 2025 - Dec 2028

National University of Singapore – School of Computing

NUS Student Life Hackathon Finalist, NUS Developer Group, IFG Squash Captain

SKILLS

Programming Languages: Python, Java, JavaScript/TypeScript, C, SQL

Technologies: React, Next.js, Node.js, Fastify, Flask, Firebase, PyTorch, Docker, SQLite, Git/GitHub

Certifications: Google Cloud Professional Machine Learning Engineer

EXPERIENCE

Digital Transformation Intern

May 2026 – August 2026

Singapore Airlines Engineering Company (SIAEC), Ground Support Equipment (GSE)

- Reduced maintenance record processing from several days to under 1 minute by building a Power Apps/Power Automate/SharePoint workflow adopted by 20 technicians and 2 supervisors
- Automated preventive maintenance scheduling for 800+ GSE by building a Power Automate workflow that sends supervisors monthly overdue and upcoming maintenance reports, replacing manual Excel tracking
- Cut daily battery-check time from 30 minutes to under 1 minute by identifying the underlying telemetry API and building a browser extension that consolidated data from 295 GSE into a searchable dashboard

Business Management Intern

January 2025 – June 2025

Aimazing (Retail Tech & Data Analytics)

- Onboarded 200+ merchants and deployed 500+ IoT loyalty devices across 3 tier-1 malls (ION Orchard, VivoCity, Wisma Atria), tracking installation and issue status via Jira
- Raised device utilization at VivoCity from 87% to 94% by authoring a 30-page merchant onboarding SOP standardizing intake, installation, and technician scheduling

PROJECTS

[mini-gabriel](#) | Python, PyTorch, Unsloth, LoRA, Slurm

Sep 2026

- Built an end-to-end pipeline from 14,939 messages: extraction, turn/session segmentation, pseudonymisation, and chat-level train/holdout splitting to prevent data leakage
- Fine-tuned Qwen3-8B with LoRA adapters covering 0.53% of model parameters on an A100 via Slurm, using response-only loss masking verified by decoding training labels before training
- Designed a calibrated style-distance metric (0.3639 to 0.0637, 83% reduction vs. baseline); quantised to 4-bit GGUF (16GB→5GB) and deployed as a CPU-only Telegram bot

[KopiKaki](#) | Gemini Live, Next.js, Firebase

August 2026

- Built a voice-first social platform for seniors using Gemini Live, Next.js, and Firebase, turning natural-language requests into confirmed local meetups
- Designed production matching infrastructure with secure Firestore rules, live candidate seeding, and Gemini function calling across meetups, people, groups, and activities

[UWash \(Finalist, Top 8 — Student Life Hackathon 2026\)](#) | Python, Flask, React, ESP32

March 2026

- Engineered a full-stack IoT laundry monitoring system using ESP32 vibration sensors and a Python Flask backend, enabling real-time machine availability tracking across residential college laundry rooms
- Developed an asynchronous Telegram bot with automated push notifications and a digital queue system, eliminating manual status updates and reducing machine idle time for UTown residential students

[Lounge Booking Mini App](#) | Node.js, Express.js, SQLite

February 2026

- Developed a Telegram Mini App for 130+ users to digitise facility bookings using Node.js, Express.js, SQLite
- Resolved a critical 64-bit integer overflow bug affecting 40% of users by enforcing strict string-casting on massive Telegram IDs before database insertion